Exactly what to do in Modeler

This is the first place you work.

1) Duplicate your old ship model

Do not touch the original file you used last semester. Save a new copy for this semester's BTP. Since all modules read the same common design/model file, keeping a clean baseline matters.

2) Check the baseline hull first

Before adding anything:

- check the stern shape
- check draft/waterline reference
- check units
- check that your original hydrostatics still match what you used before

Modeler is intended for hull fairness and direct manipulation of NURBS surfaces, and it provides upright hydrostatics and curve-of-areas checks right in the design environment.

3) Add the propeller placeholder as a separate surface

Do **not** merge it with the hull.

Because I cannot see your exact version/build, I do not want to invent ribbon names. In Modeler, look for the commands corresponding to:

- creating a **new NURBS surface**
- **scaling** a surface
- **rotating** a surface
- **translating/moving** a surface
- optionally **grouping/masking** surfaces

Those capabilities are explicitly part of Modeler: unlimited NURBS surfaces, interactive control-point movement, trimming, grouping, masking, and parametric transformations.

What to actually make:

- a small circular or elliptical **disc**
- centered on the vessel centerline for a single-screw vessel, so usually yp=0y_p = 0yp=0
- located aft of the stern at the propeller center
- oriented normal to the shaft axis

Name that surface something obvious like:

`PROP_REF_DISC`

Put it in its own group or layer.

4) Record the propeller reference numbers

You need these values from Modeler/design setup:

- xp,yp,zpx_p, y_p, z_pxp,yp,zp: propeller-center coordinates
- DDD: propeller diameter
- shaft-axis direction
- distance from calm-water line to propeller center

Then compute the still-water depth of the **top** of the propeller:

htop,static=(depth of propeller center below waterline)−D2h_{\text{top,static}} = (\text{depth of propeller center below waterline}) - \frac{D}{2}htop,static=(depth of propeller center below waterline)−2D

This one number becomes the basis of your emergence label later.

5) Keep the disc out of the real hydrostatic envelope

This is critical.

If your version lets you choose surfaces for hydrostatics/stability analysis, exclude `PROP_REF_DISC`. Maxsurf's public material explicitly says a single model can contain mixed geometry and selected surfaces can be chosen for analysis; Stability also defines the hydrostatic envelope from the hull/superstructure/appendage surfaces.

Your test for correctness is simple:

- run the baseline hydrostatics before adding the disc
- run them again after adding the disc
- if displacement/hydrostatics changed, the disc is being treated incorrectly

Exactly what to do in Stability

For your BTP, Stability is **not** where you simulate the propeller.
It is where you define the **operating condition** of the ship.

Stability supports loadcases, tanks, compartments, upright hydrostatics, equilibrium, large-angle stability, and related analyses. Loadcases are the right way to define mass and CG conditions.

Use Stability for these tasks only

1) Define load groups / load cases

At minimum define:

- lightship
- one or more loading conditions
- fuel / cargo / tank assumptions only if relevant to your vessel

Stability's loadcase system is designed for exactly this purpose.

2) Run upright hydrostatics / equilibrium

For each loading condition, get:

- displacement
- draft
- trim
- LCG / VCG / maybe GM if needed
- stern draft or local immersion-related quantity if available in your workflow

These are useful as **inputs** to Motions and later to ML. Stability explicitly supports upright hydrostatics and equilibrium analyses.

3) Ignore damage and large-angle stability unless your professor asks

For this BTP, you usually do **not** need:

- damage cases
- floodable length
- probabilistic damage
- code compliance reports

Those are real Stability capabilities, but they are not central to your propeller-operability ML workflow.

Exactly what to do in Motions

This is your main simulation module.

Motions reads the hull geometry directly from the trimmed Maxsurf model and computes RAOs, response spectra, added resistance, significant absolute and relative motions, velocities, and accelerations for specified wave spectra, headings, and speeds. It also lets you compute response at remote points on the vessel.

1) Open the same design/model file

Do not export offsets manually. Motions is meant to read directly from the common Maxsurf model.

2) For each case, set the operating condition

For each run, specify:

- loading condition from Stability
- vessel speed
- wave heading
- wave spectrum / sea-state parameters
- any other analysis parameters your setup requires

That is how Motions is documented to work.

3) Add a remote output point at the propeller center

This is the whole point of the disc.

Since Motions can compute motion, velocity, and acceleration at **any position on the vessel**, define a point at:

(xp,yp,zp)(x_p, y_p, z_p)(xp,yp,zp)

which is the propeller center you fixed in Modeler.

4) Export these outputs from Motions

Do not export everything. Export only what you need.

Save:

- your original 6 motion-response quantities
- added resistance
- vertical motion at propeller center
- vertical acceleration at propeller center
- possibly relative motion at that location if your workflow exposes it

Also keep the case metadata:

- speed
- heading
- loading condition
- sea-state parameters

Motions explicitly supports copying data to spreadsheets and other applications, and its own docs specifically mention doing further analysis such as **propeller emergence** outside the module.

Exactly what to do for resistance / power

Depending on your version, do this either in **Resistance** or in **Motions** if your release includes the 2025 integration of calm-water resistance algorithms.

Export:

- total calm-water resistance
- effective power / required power
- wake-related value if available in your chosen method

Resistance is explicitly the Maxsurf module for resistance, power requirements, and wake calculations.

What you should compute outside Maxsurf

This is where the propeller "analysis" really happens.

Use the motion at the propeller center to define an emergence score. Since Maxsurf's own docs say Motions data can be exported for further analysis such as propeller emergence, this step belongs outside the solver.

A simple and very usable label is:

E=Aupward at prop centerhtop,staticE = \frac{A_{\text{upward at prop center}}}{h_{\text{top,static}}}E=htop,staticAupward at prop center

where:

- Aupward at prop centerA_{\text{upward at prop center}}Aupward at prop center = upward motion amplitude or relevant vertical-response measure at the propeller point
- htop,statich_{\text{top,static}}htop,static = static depth of propeller top below the waterline

Then define:

- **Emergence flag** = 1 if E≥1E \ge 1E≥1
- **Emergence flag** = 0 if E<1E < 1E<1

Or keep EEE itself as a continuous severity score.

What the ML should predict

Do **not** try to predict 20 things in your first model.

The cleanest target set is:

Must-have targets

- future / next-case **6 motion responses**
- **propeller emergence score** EEE
- **propeller emergence flag** 0/1
- **added resistance**

Good extra targets

- **calm-water resistance**
- **required/effective power**
- **vertical motion at propeller center**
- **vertical acceleration at propeller center**

What should be inputs

Use these as inputs to ML:

- draft / displacement / trim from Stability
- speed
- wave heading
- wave spectrum / sea-state parameters
- propeller-center coordinates (xp,yp,zp)(x_p, y_p, z_p)(xp,yp,zp)
- propeller diameter DDD
- optionally some principal particulars or hydrostatic values

Which ML models make sense

If your data is **case-wise** rather than long time-series:

- XGBoost
- Random Forest
- LightGBM
- MLP

If you really have **time-sequence data** like last semester:

- LSTM
- GRU
- TCN
- time-series Transformer

My recommendation is to start with this exact prediction package:

1. regress the 6 motion outputs
2. classify emergence flag
3. regress resistance and power

That gives you one nice BTP story:
ship motions + propeller operability + propulsion-related performance

The shortest exact workflow

In Modeler / design file

- keep old hull
- add separate `PROP_REF_DISC`
- record xp,yp,zp,Dx_p,y_p,z_p,Dxp,yp,zp,D
- make sure hull hydrostatics do not change

In Stability

- define loadcases
- run upright hydrostatics / equilibrium
- export draft, displacement, trim, KG/GM-type condition data

In Motions

- run sea-state / heading / speed cases
- define propeller-center output point
- export 6 motions + point response + added resistance

In Resistance or Motions

- export calm-water resistance + power + wake if available

Outside Maxsurf

- compute emergence score / flag
- train ML models

The one thing I would not do is treat the disc as a real hydrodynamic propeller model. In your project, it should remain a **reference geometry** for extracting local stern-response and operability metrics.